

WUNSTORF, Germany

WMO: 103340

Lat: 52.4606N

Lon: 9.4311E

Elev: 69

StdP: 100.50

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.2	-6.9	-12.0	1.3	-8.0	-9.9	1.6	-5.9	12.7	7.5	11.4	6.8	3.7	80	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	30.9	19.6	28.5	18.8	26.4	18.0	20.6	28.3	19.7	26.6	18.8	24.8	3.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.1	22.5	17.1	12.3	22.0	16.2	11.6	21.3	59.4	28.5	56.3	26.7	53.4	24.9	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.2	9.0	8.0	-11.8	34.5	3.4	1.8	-14.3	35.7	-16.3	36.8	-18.3	37.8	-20.8	39.1		
			-11.9	22.6	3.5	1.0	-14.5	23.3	-16.5	23.9	-18.5	24.4	-21.1	25.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	2.0	2.8	5.5	9.8	13.7	16.8	19.2	18.9	15.1	10.7	6.1	3.0
	DBStd	7.21	4.71	4.43	3.81	3.86	3.65	3.40	3.35	3.11	3.07	3.61	3.65	4.61
	HDD10.0	1034	250	203	144	51	7	0	0	0	1	36	124	218
	HDD18.3	3058	507	436	397	255	150	69	28	30	107	237	367	474
	CDD10.0	1162	1	1	6	46	123	204	287	276	153	57	7	2
	CDD18.3	144	0	0	0	0	7	24	57	48	9	0	0	0
	CDH23.3	1349	0	0	0	9	83	229	536	417	72	3	0	0
	CDH26.7	422	0	0	0	1	15	62	187	144	14	0	0	0
(14) Wind	WSAvg	3.9	4.6	4.4	4.4	3.7	3.5	3.5	3.5	3.3	3.4	3.8	4.0	4.4
(15) Precipitation	PrecAvg	785	72	55	52	48	65	66	81	78	65	66	68	71
	PrecMax	1053	125	121	126	91	160	116	186	185	148	186	133	169
	PrecMin	599	14	17	9	8	23	26	30	23	17	21	8	23
	PrecStd	116	32	27	30	21	34	23	36	36	32	32	30	30
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	14.2	18.9	24.8	28.8	31.8	34.0	33.2	28.8	22.6	15.9	13.2
		MCWB	10.0	8.9	11.0	15.0	18.5	20.9	20.3	20.4	19.1	16.5	13.0	10.6
	2%	DB	11.1	11.9	15.6	21.7	25.7	28.2	31.1	30.4	25.3	19.3	14.0	12.0
		MCWB	8.8	8.4	10.0	13.4	16.8	19.0	19.7	19.2	17.6	15.0	11.5	9.8
	5%	DB	9.4	10.2	13.1	19.2	23.2	26.0	28.7	27.8	22.8	17.4	12.8	10.2
		MCWB	7.7	7.7	9.0	12.2	15.2	17.8	18.9	18.6	16.5	14.0	10.6	8.3
	10%	DB	8.0	8.9	11.2	17.0	21.0	23.8	26.2	25.2	20.4	15.9	11.2	9.0
		MCWB	6.4	6.8	8.0	11.3	14.1	16.8	18.1	17.6	15.5	13.3	9.1	7.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	10.2	12.3	15.8	19.9	21.5	21.8	21.5	19.6	16.9	13.4	11.3
		MCDB	12.4	12.4	16.7	22.2	27.1	28.9	29.6	29.7	26.2	21.3	15.0	13.0
	2%	WB	9.2	9.1	10.8	14.2	17.7	19.8	20.7	20.3	18.4	15.5	11.8	10.1
		MCDB	10.9	11.3	14.3	20.0	23.8	26.5	28.4	27.8	24.0	18.8	13.7	11.7
	5%	WB	7.7	7.9	9.5	13.0	16.1	18.6	19.8	19.3	17.1	14.4	10.7	8.6
		MCDB	9.2	10.1	12.5	17.9	22.0	24.6	26.8	25.8	21.5	16.9	12.5	10.1
	10%	WB	6.4	6.7	8.3	11.8	14.8	17.4	18.7	18.4	16.1	13.3	9.5	7.3
		MCDB	7.8	8.7	11.0	16.5	19.6	22.5	24.9	24.0	19.9	15.6	11.0	8.7
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.6	7.3	9.5	10.0	9.9	10.0	10.0	8.8	7.2	4.9	4.4
		MCDBR	5.8	7.3	11.1	13.5	13.9	13.8	14.3	14.4	12.5	9.0	6.3	5.3
	5% WB	MCWBR	4.7	5.1	6.7	7.0	6.9	6.4	5.6	5.8	6.2	5.6	4.7	4.5
		MCDWR	5.3	6.3	9.6	11.9	12.4	12.3	12.6	11.9	10.8	8.5	5.9	5.1
(40) Clear-Sky Solar Irradiance	taub	0.319	0.340	0.374	0.411	0.412	0.410	0.425	0.418	0.387	0.357	0.337	0.314	0.314
		taud	2.364	2.351	2.272	2.214	2.247	2.272	2.241	2.276	2.355	2.420	2.392	2.355
	Ebn at Noon	670	766	807	821	840	843	821	805	788	738	640	604	604
		Edh at Noon	60	81	106	127	129	127	118	97	75	59	52	52
(44) All-Sky Solar Radiation	RadAvg	0.63	1.32	2.38	3.93	4.81	5.14	4.92	4.13	3.01	1.73	0.78	0.50	0.50
	RadStd	0.06	0.19	0.27	0.47	0.52	0.39	0.65	0.31	0.37	0.16	0.09	0.05	0.05

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.40	N/A	N/A	N/A	N/A	+0.48	N/A	-160	N/A	N/A
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-124	N/A	N/A

Nomenclature: See separate page